

INTER QUARTILE RANGE PLACEMENT DATASET

IQR CALCULATION:

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.303395	66.333163	66.370186	72.100558	62.278186	288655.405405
Median	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0
Q1:25%	54.5	60.6	60.9	61.0	60.0	57.945	240000.0
Q2:50%	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Q3:75%	161.5	75.7	73.0	72.0	83.5	66.255	300000.0
99%	212.86	87.0	91.86	83.86	97.0	76.1142	NaN
Q4:100%	215.0	89.4	97.7	91.0	98.0	77.89	940000.0
IQR	107.0	15.1	12.1	11.0	23.5	8.31	60000.0
1.5_Rule	160.5	22.65	18.15	16.5	35.25	12.465	90000.0
Lesser_range	-106.0	37.95	42.75	44.5	24.75	45.48	150000.0
Greater_range	1.0	53.05	54.85	55.5	48.25	53.79	210000.0
Min	1	40.89	37.0	50.0	50.0	51.21	200000.0
Max	215	89.4	97.7	91.0	98.0	77.89	940000.0

1. Ssc_p Column Calculation:

Min = 40.89, Max = 89.4

IQR = Q3-Q1 = 75.7-60.6 = 15.1

Lesser range = Q1-1.5*IQR

= 60.6 – (1.5*15.1)

= 37.95

Greater range = Q3+1.5*IQR

= 75.7+(1.5*15.1)

= 98.35

So, there is no lesser and greater outlier in ssc_p column.

2. Hsc_p Column Calculation:

Min = 37, Max = 97.7

$$\text{IQR} = \text{Q3} - \text{Q1} = 73 - 60.9 = 12.1$$

$$\text{Lesser range} = \text{Q1} - 1.5 * \text{IQR}$$

$$= 60.9 - (1.5 * 12.1)$$

$$= 42.75$$

$$\text{Greater range} = \text{Q3} + 1.5 * \text{IQR}$$

$$= 73 + (1.5 * 12.1)$$

$$= 91.15$$

So, there is lesser outlier and no greater outlier in hsc_p column.

3. Degree_p Column Calculation:

Min = 50, Max = 91

$$\text{IQR} = \text{Q3} - \text{Q1} = 72 - 61 = 11$$

$$\text{Lesser range} = \text{Q1} - 1.5 * \text{IQR}$$

$$= 61 - (1.5 * 11)$$

$$= 44.5$$

$$\text{Greater range} = \text{Q3} + 1.5 * \text{IQR}$$

$$= 72 + (1.5 * 11)$$

$$= 88.5$$

So, there is no lesser outlier but there is greater outlier in degree_p column.

4. Etest_p Column Calculation:

Min = 50, Max = 98

$$\text{IQR} = \text{Q3} - \text{Q1} = 83.5 - 60 = 23.5$$

$$\text{Lesser range} = \text{Q1} - 1.5 * \text{IQR}$$

$$= 60 - (1.5 * 23.5)$$

$$= 24.75$$

$$\text{Greater range} = \text{Q3} + 1.5 * \text{IQR}$$

$$= 83.5 + (1.5 * 23.5)$$

$$= 118.75$$

So, there is no lesser outlier and greater outlier in etest_p column.

5. Etest_p Column Calculation:

Min = 50, Max = 98

$$\text{IQR} = \text{Q3} - \text{Q1} = 83.5 - 60 = 23.5$$

$$\text{Lesser range} = \text{Q1} - 1.5 * \text{IQR}$$

$$= 60 - (1.5 * 23.5)$$

$$= 24.75$$

$$\text{Greater range} = \text{Q3} + 1.5 * \text{IQR}$$

$$= 83.5 + (1.5 * 23.5)$$

$$= 118.75$$

So, there is no lesser outlier and greater outlier in etest_p column.

6. Salary Column Calculation:

Min = 200000, Max = 940000

$$\text{IQR} = \text{Q3} - \text{Q1} = 300000 - 240000 = 60000$$

$$\text{Lesser range} = \text{Q1} - 1.5 * \text{IQR}$$

$$= 240000 - (1.5 * 60000)$$

$$= 150000$$

$$\text{Greater range} = \text{Q3} + 1.5 * \text{IQR}$$

$$= 300000 + (1.5 * 60000)$$

$$= 390000$$

So, there is no lesser outlier but there is greater outlier in salary column.